

§4.2

Events and Sample Spaces

List every outcome, or count without listing

Today's Goal

- List the outcomes in a sample space using various methods
- Use tree diagrams to enumerate outcomes
- Identify simple and compound events
- Find the complement of an event
- Apply counting techniques to determine the size of a sample space

What You Will Be Graded On

LT.7 Probability

I understand what the probability of an event represents and can apply basic rules of probability to calculate the probability of an event.

Two Four-Sided Dice

Two fair four-sided dice, each numbered 1 through 4, are rolled.

Which sum of the two numbers is most likely?

Listing in a Steady Order

For simple experiments, we can list all outcomes directly. For more complex experiments, we need systematic methods.

List the sample space for answering three true/false questions.

We can systematically list all possible sequences of true (T) and false (F):

$$S = \{(T, T, T), (T, T, F), (T, F, T), (T, F, F), \\ (F, T, T), (F, T, F), (F, F, T), (F, F, F)\}.$$

There are 8 possible outcomes.

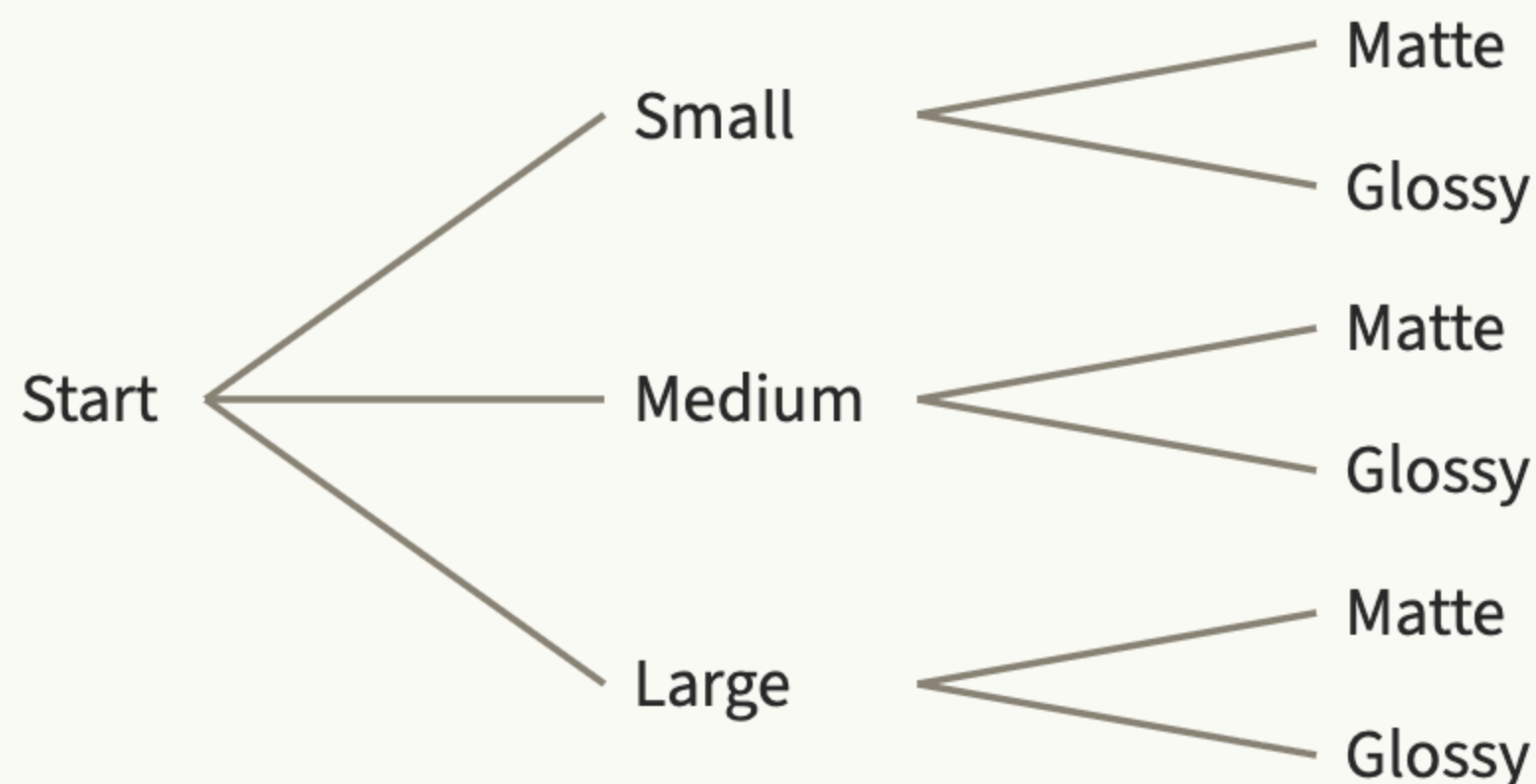
Branch at Each Stage

Definition 4.2.2

A *tree diagram* is a visual tool that systematically lists all possible outcomes of a multi-stage probability experiment by branching at each stage. Each complete root-to-leaf path corresponds to one outcome in the sample space.

One Path, One Outcome

A print shop offers a poster in a choice of size (small, medium, or large) and a choice of finish (matte or glossy). Use a tree diagram to list all possible posters.



Reading from left to right along each path, there are 6 possible posters.

A Table for Two Stages

When an experiment has exactly two stages, a *table* can be an efficient way to list all outcomes.

Two fair four-sided dice are rolled. Use a table to list all possible outcomes.

	1	2	3	4
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)

The sample space contains $4 \times 4 = 16$ outcomes.

So Which Sum Is Most Likely?

	1	2	3	4
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)

The event "the sum is 5" is $\{(1, 4), (2, 3), (3, 2), (4, 1)\}$, a diagonal of the table.

$$P(\text{sum is 5}) = \frac{4}{16} = 0.25.$$

No other sum has as many as 4 outcomes. The sums 2 and 8 have only 1 each.

Count Without Listing

Theorem 4.2.6 Fundamental Counting Principle

If one event can occur in m ways and a second event can occur in n ways (after the first event has occurred), then the two events can occur together in $m \cdot n$ ways. More generally, if k events can occur in $n_1, n_2, \dots, n_k$ ways respectively, then the total number of ways all k events can occur is $n_1 \cdot n_2 \cdot \dots \cdot n_k$.

Multiply the Choices

A locker code consists of 2 letters followed by 3 digits. How many different locker codes are possible?

- There are 26 choices for each letter position.
- There are 10 choices (0–9) for each digit position.

By the Fundamental Counting Principle:

$$26 \times 26 \times 10 \times 10 \times 10 = 26^2 \times 10^3 = 676,000.$$

There are 676,000 possible locker codes.

Now You Try: PIN Codes

You Try 4.2.3. A PIN code consists of 4 digits. How many different PIN codes are possible if:

(a) Digits can be repeated?

$$10 \times 10 \times 10 \times 10 = 10,000$$

(b) Digits cannot be repeated?

$$10 \times 9 \times 8 \times 7 = 5,040$$

One Outcome, or More Than One

Definition 4.2.8

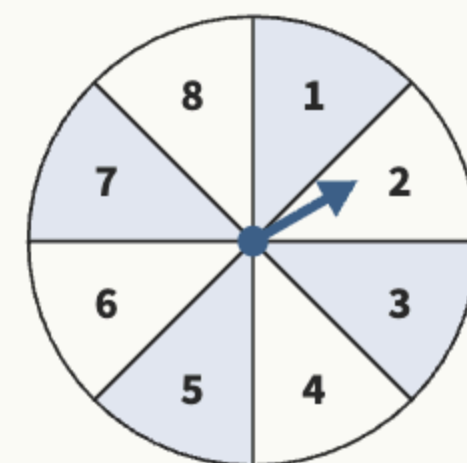
A *simple event* is an event that consists of exactly one outcome from the sample space.

Definition 4.2.9

A *compound event* is an event that consists of two or more outcomes (simple events).

Simple or Compound?

A spinner with 8 equal sections numbered 1 through 8 is spun. Classify each event as simple or compound:



(a) Event A : Spinning a 6.

Simple event. Event $A = \{6\}$ contains exactly one outcome.

(b) Event B : Spinning an even number.

Compound event. Event $B = \{2, 4, 6, 8\}$ contains four outcomes.

(c) Event C : Spinning a number greater than 5.

Compound event. Event $C = \{6, 7, 8\}$ contains three outcomes.

Everything That Is Not E

Definition 4.2.11

The *complement* of an event E , denoted E' , is the event consisting of all outcomes in the sample space that are NOT in E .

The probability of an event and its complement always sum to 1:

$$P(E') = 1 - P(E) \quad \text{or equivalently} \quad P(E) + P(E') = 1.$$

Two Ways to Reach “Not”

A jar contains 4 red marbles, 6 blue marbles, and 10 green marbles. A marble is randomly selected. Find the probability of NOT selecting a blue marble.

Let E = “selecting a blue marble.” There are $4 + 6 + 10 = 20$ marbles total:

$$P(E) = \frac{6}{20} = 0.3.$$

Using the complement: $P(E') = 1 - P(E) = 1 - 0.3 = 0.7$.

Alternatively, we could count directly: $P(E') = \frac{4+10}{20} = \frac{14}{20} = 0.7$.

At Least One Head

A fair coin is flipped four times. We want the probability that at least one flip is heads.

**How many outcomes
would you have to count?**

Count the Complement Instead

The complement is especially useful when calculating $P(E)$ directly is difficult, but calculating $P(E')$ is easier.

Let E = “at least one head.” The complement $E' =$ “no heads” = “all four are tails.”

The sample space for four flips has $2^4 = 16$ equally likely outcomes.

There is only 1 way to have all tails (T, T, T, T) , so $P(E') = P(\text{all tails}) = \frac{1}{16}$.

Using the complement: $P(E) = 1 - P(E') = 1 - \frac{1}{16} = \frac{15}{16} = 0.9375$.

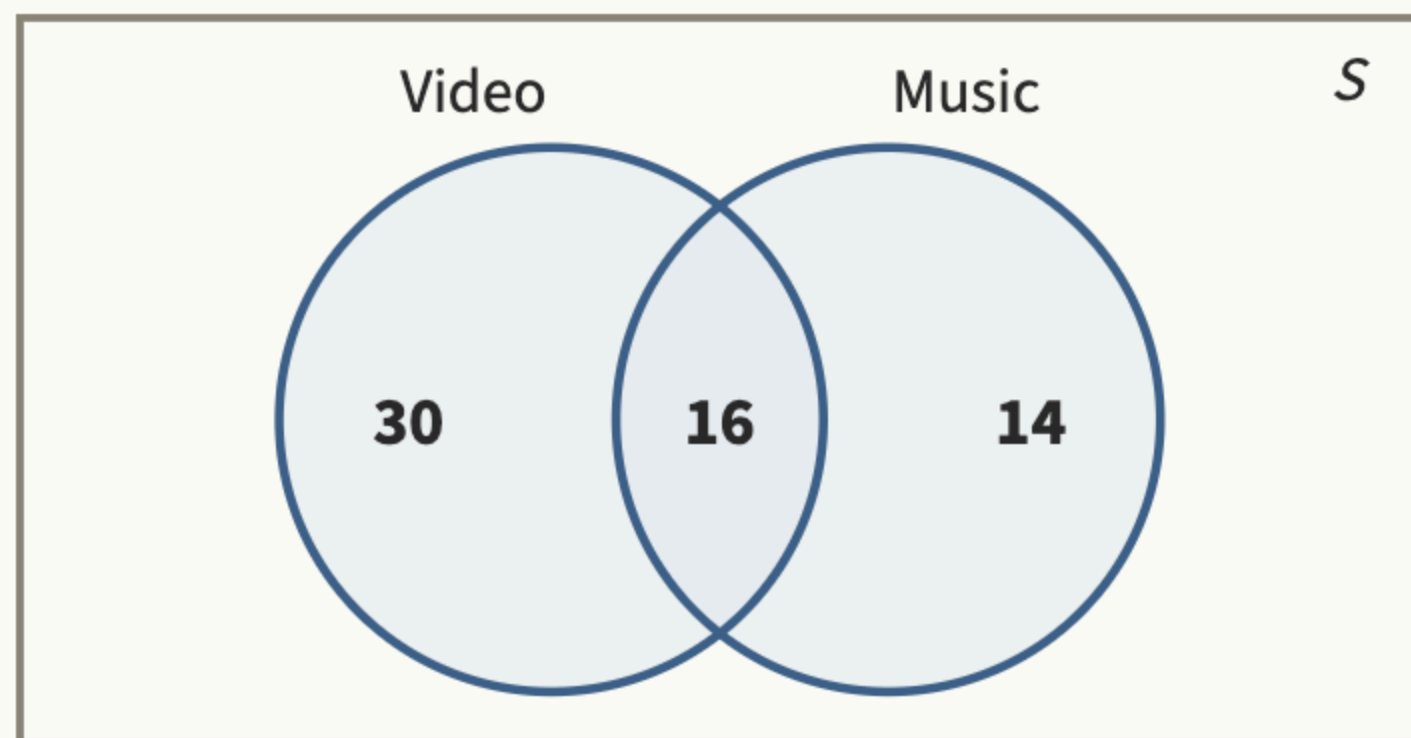
A Picture of Events

Definition 4.2.15

A *Venn diagram* is a visual representation of events and their relationships within a sample space. The sample space is represented by a rectangle, and events are represented by circles or ovals within the rectangle. Overlapping regions represent intersections of events; non-overlapping regions represent mutually exclusive parts.

Fill In From the Overlap

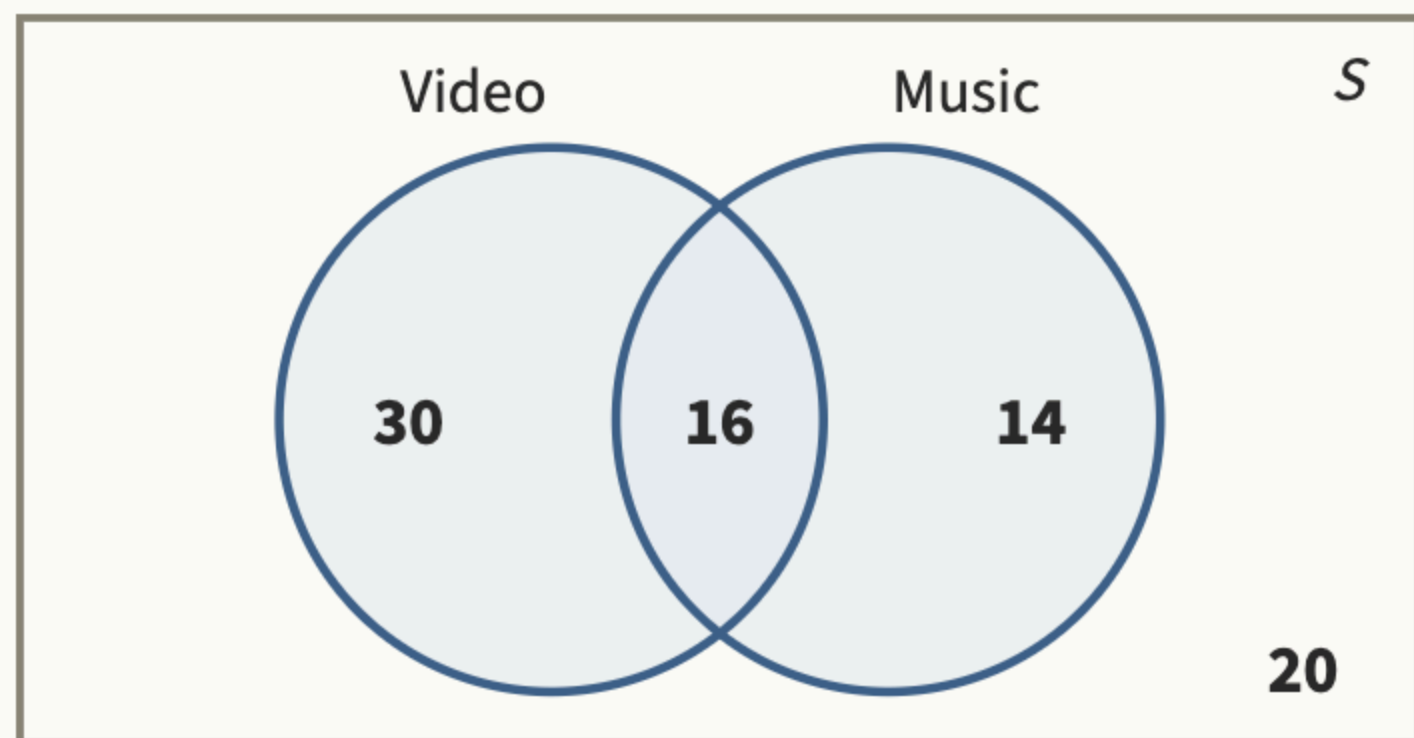
In a survey of 80 students, 46 have a video subscription, 30 have a music subscription, and 16 have both. Draw a Venn diagram representing this situation.



- 16 students have both subscriptions (the intersection).
- $46 - 16 = 30$ students have only a video subscription.
- $30 - 16 = 14$ students have only a music subscription.

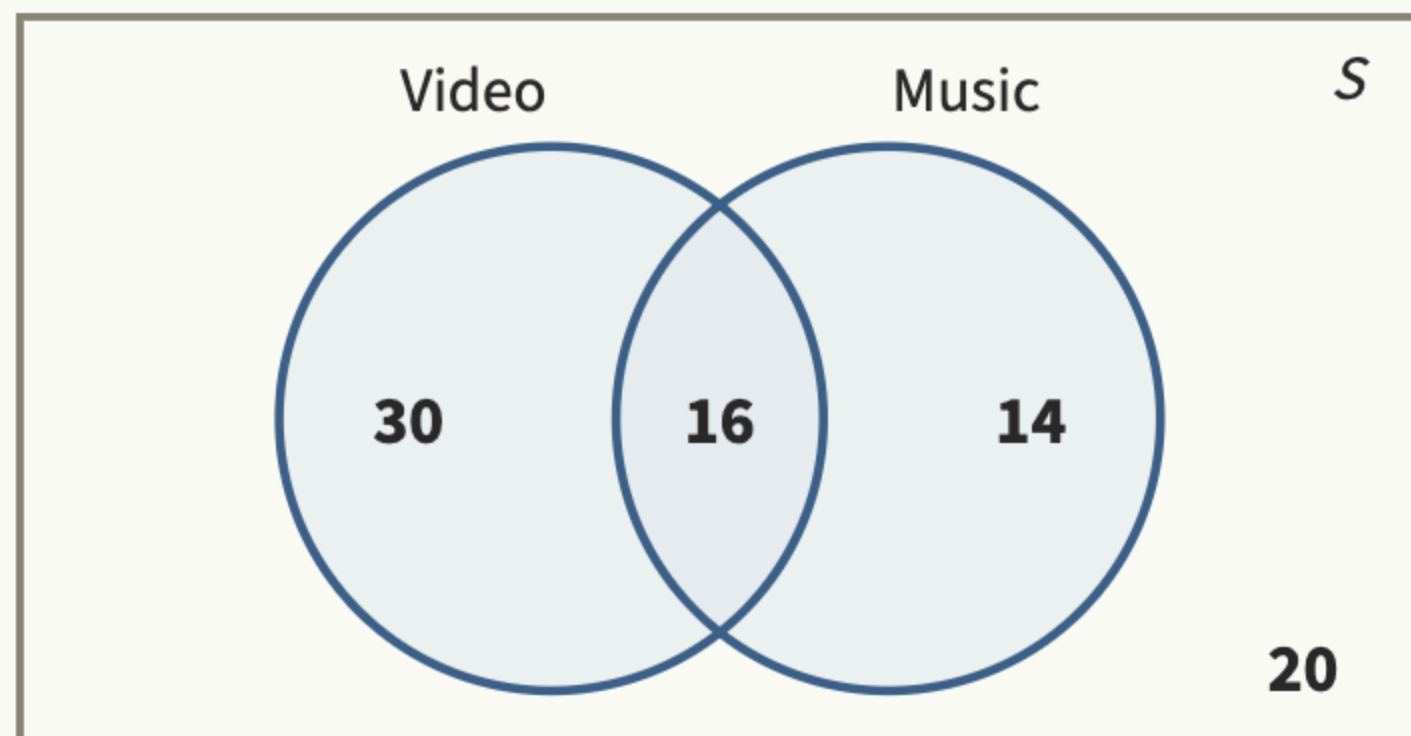
Which Region Is Still Missing?

The three regions inside the circles account for $30 + 16 + 14 = 60$ of the 80 students.



$80 - 30 - 16 - 14 = 20$ students have neither subscription. They are counted inside S but outside both circles.

Probabilities From the Diagram



If a student is randomly selected, find the probability that the student:

(a) has a video subscription.

$$P(\text{video}) = \frac{46}{80} = 0.575.$$

(b) has neither subscription.

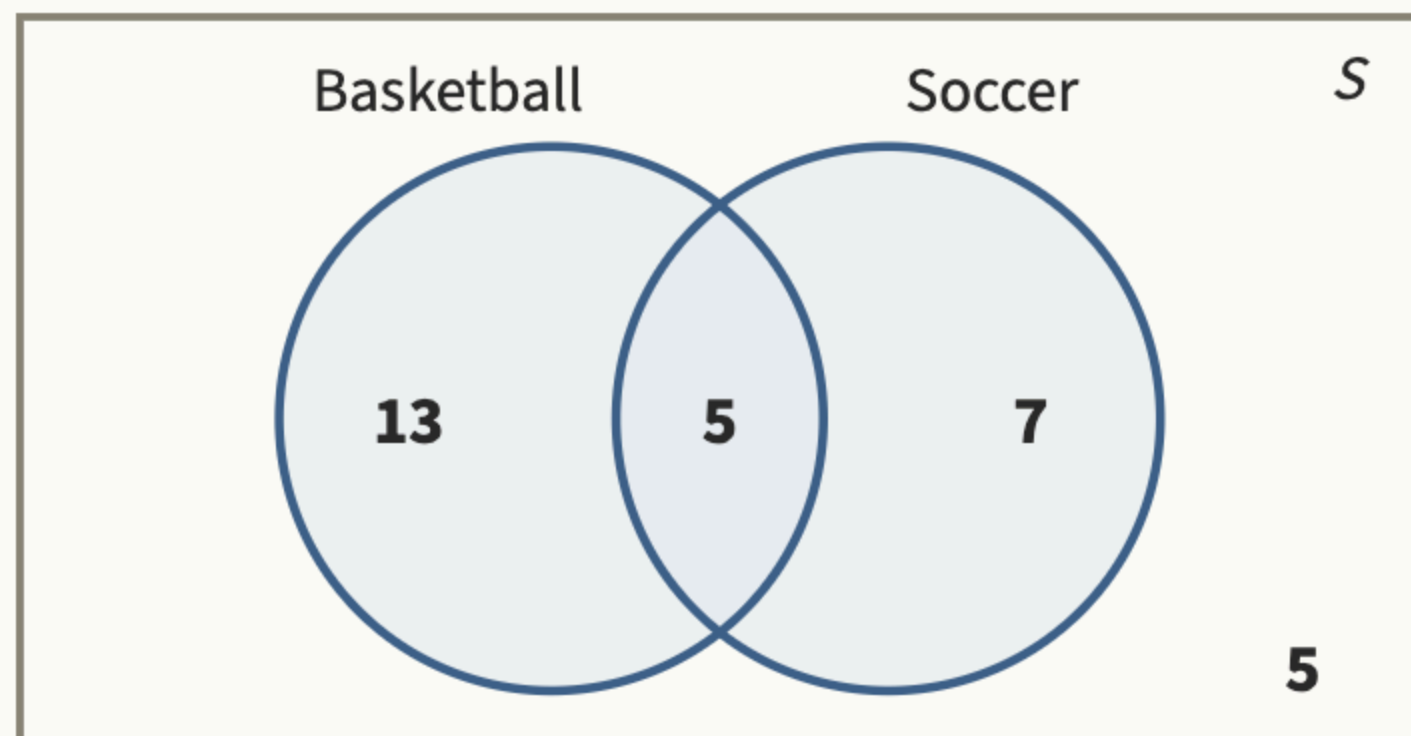
$$P(\text{neither}) = \frac{20}{80} = 0.25.$$

Now You Try: Two Sports

You Try 4.2.5. In a class of 30 students, 18 play basketball, 12 play soccer, and 5 play both sports. Draw a Venn diagram and determine:

- (a) How many students play only basketball?
- (b) How many students play only soccer?
- (c) How many students play neither sport?
- (d) If a student is randomly selected, what is the probability that the student plays at least one sport? Round to 4 decimal places.

You Try 4.2.5: Checking Your Answers



(a) Only basketball

$$18 - 5 = 13$$

(b) Only soccer

$$12 - 5 = 7$$

(c) Neither sport

$$30 - 13 - 5 - 7 = 5$$

(d) At least one sport

$$P(\text{at least one}) = \frac{25}{30} \approx 0.8333$$

What to Take With You

- List, tree, or table: each one shows every outcome. A table only works for two stages.
- The Fundamental Counting Principle gives the count without the list. Watch for “after the first event has occurred.”
- $P(E') = 1 - P(E)$. “At least one” complements to “none.”
- In a Venn diagram, start from the overlap, and do not forget “neither.”